Assignment 5

1. Consider the action of the group of upper triangular matrices in $GL_n(\mathbb{R})$ (non-zero entries on the diagonal allowed) acting on $\mathbb{R}^n - \{0\}$. Is this action :
   a) Faithful ? b) Free ? c) Transitive ?

2. Show that if p is a prime and G is a group of order p^n, then for every $m < n$, G has a subgroup of order p^m.

3. Show that $\mathbb{Z}$ is not isomorphic to $\mathbb{Z} \oplus \mathbb{Z}$.

4. Let G be a group of order pn where p is a prime number and $p > n$. Show that if H is a subgroup of order p then it is a normal subgroup of G.

5. Prove that a group of order 56 has a normal Sylow p-subgroup for some prime p dividing 56.